REGULARITY DESCENDS FROM THE COMPOSITION ALONG A FLAT MAP SURJECTIVE ON SPECTRA

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- Flatness is permanent in the composition. Algebra, Lemma \ref{algebra-lemma-flat-permanence}, cited at proof lines 10513-10514. Gives flatness of the first map.
- Geometric regularity descends along a flat map surjective on spectra. Algebra, Lemma \ref{algebra-lemma-geometrically-regular-descent}, cited at proof lines 10517-10519. Gives geometric regularity of the fibre.

Source reference. Original lemma 07NT: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 9773–9773. The lemma's own canonical tag, quoted so that readers can resolve the label inside the delivered text; it points at the counted statement itself and carries no proof content.

Source reference. Original lemma 039V: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 4286–4286. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 07NH: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 9762–9762. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Lemma 0.2. *Let $A \to B \to C$ be ring maps. If $A \to C$ is regular and $B \to C$ is flat and surjective on spectra, then $A \to B$ is regular.*

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Proof. By Algebra, Lemma 039V we see that $A \to B$ is flat. Let $\mathfrak{p} \subset A$ be a prime. The ring map $B \otimes_A \kappa(\mathfrak{p}) \to C \otimes_A \kappa(\mathfrak{p})$ is flat and surjective on spectra. Hence $B \otimes_A \kappa(\mathfrak{p})$ is geometrically regular by Algebra, Lemma 07NH. $\square$